

E-Commerce Sales & Customer Analytics

Project Explanation Report | Het Savani

E-COMMERCE SALES & CUSTOMER ANALYTICS - PROJECT EXPLANATION

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Tools: Python, MySQL, Power BI, Streamlit, Plotly

1. PROJECT OVERVIEW

This is an end-to-end data analytics project that simulates a real-world e-commerce business scenario. The goal is to analyze sales performance, customer behavior, and product trends using a complete data pipeline - from raw data generation to an interactive business dashboard.

This type of project is used by companies to answer questions like:

- * How much revenue did we make this month vs last month?
- * Which products and categories are selling the most?
- * Which customers are loyal and which are at risk of leaving?
- * Which cities generate the most revenue?
- * How do customers prefer to pay?

2. PROBLEM STATEMENT

A business has thousands of orders, customers, and products.

The raw data is spread across multiple tables and is difficult to read or understand directly. We need to:

- Organize the data into a clean database structure
- Clean and process the raw data using Python
- Create analytics summaries using SQL
- Visualize insights in an interactive dashboard
- Segment customers to identify high-value vs at-risk groups

3. TOOLS & TECHNOLOGIES USED

PYTHON (version 3.12)

- * Used for: data generation, cleaning, and loading
- * Libraries: mysql-connector-python, pandas, streamlit, plotly
- * Why Python: industry standard for data engineering and analytics

MYSQL (version 9.7)

- * Used for: storing and organizing the data
- * Why MySQL: most widely used open-source relational database
- * We used SQL views to pre-calculate analytics summaries

POWER BI

- * Used for: building the interactive business dashboard
- * Connected to MySQL via ODBC driver
- * Why Power BI: industry standard business intelligence tool

STREAMLIT + PLOTLY

- * Used for: building a web-based analytics dashboard
- * Why: lightweight, Python-native, runs in browser instantly

4. PROJECT STRUCTURE

```
ecommerce_analytics/
|
+-- config.py          ? Database connection settings
+-- setup.py           ? Master script: runs everything
+-- dashboard.py       ? Streamlit web dashboard
|
+-- scripts/
|   +-- generate_data.py ? Creates realistic fake data
|   +-- load_data.py     ? Cleans and loads data to MySQL
|
+-- sql/
|   +-- schema.sql       ? Creates database tables
|   +-- views.sql        ? Creates analytics views
|
+-- data/
+-- raw/                ? Generated CSV files stored here
```

5. DATABASE DESIGN (Star Schema)

We designed a normalized relational database with 5 tables:

CUSTOMERS

- * customer_id (primary key)
- * customer_name, email, city, state, country
- * registration_date
- * Total: 2,000 customers

PRODUCTS

- * product_id (primary key)
- * product_name, category, price, cost
- * 8 categories: Electronics, Clothing, Books, Sports, Home & Garden, Toys, Beauty, Food & Grocery
- * Total: 48 products

ORDERS

- * order_id (primary key)
- * customer_id (foreign key ? customers)
- * order_date, status (delivered/cancelled/returned)
- * Total: 15,000 orders

ORDER_ITEMS

- * order_item_id (primary key)
- * order_id (foreign key ? orders)

- * product_id (foreign key ? products)
- * quantity, unit_price, discount
- * Total: 44,816 order items

PAYMENTS

- * payment_id (primary key)
- * order_id (foreign key ? orders)
- * payment_date, amount, method
- * Methods: Credit Card, Debit Card, UPI, Net Banking, COD
- * Total: 11,879 payments

WHY THIS DESIGN?

- * Avoids data duplication
- * Each piece of information is stored only once
- * Tables are linked through foreign keys
- * This is called a "Star Schema" - industry standard for data warehousing and analytics

6. DATA PIPELINE (ETL Process)

ETL stands for Extract, Transform, Load - the standard process for moving data from source to database.

STEP 1 - EXTRACT (generate_data.py)

- * Generated 2,000 realistic customers across 8 Indian cities
- * Generated 48 products across 8 categories with real prices
- * Generated 15,000 orders over 3 years (2022-2024)
- * Generated 44,816 order items with quantities and discounts
- * Saved all data as CSV files

STEP 2 - TRANSFORM (load_data.py)

- * Cleaned customer names (proper capitalization)
- * Standardized email addresses (lowercase)
- * Converted data types (strings to numbers where needed)
- * Removed duplicates using INSERT IGNORE

STEP 3 - LOAD (load_data.py)

- * Connected to MySQL database
- * Cleared existing data to avoid duplicates
- * Loaded all cleaned CSV data into MySQL tables
- * Used batch insert for performance (faster than row-by-row)

7. SQL ANALYTICS VIEWS

We created 7 SQL views - these are pre-written queries saved in the database that can be reused by any tool (Power BI, Python).

VIEW 1: vw_monthly_revenue

- * Shows revenue, orders, and customers per month
- * Used for: revenue trend analysis
- * Key insight: track growth or decline month by month

VIEW 2: vw_category_performance

- * Shows revenue, units sold, and profit per product category
- * Used for: identifying best and worst performing categories
- * Key insight: Electronics is the top revenue category

VIEW 3: vw_top_products

- * Shows best selling products ranked by revenue
- * Used for: product performance analysis
- * Key insight: Smartphone X12 and Wireless Earbuds are top sellers

VIEW 4: vw_customer_cohorts

- * Groups customers by the month they first purchased
- * Tracks how much each group spends over time
- * Used for: customer retention analysis
- * Key insight: identifies which months acquired most valuable customers

VIEW 5: vw_rfm

- * This is the most advanced analytics view ***

RFM stands for Recency, Frequency, Monetary

- * RECENCY: how recently did the customer buy? (lower = better)
- * FREQUENCY: how many times did they buy? (higher = better)
- * MONETARY: how much did they spend? (higher = better)

Each customer gets a score 1-5 on each dimension.

Then they are grouped into segments:

- * Champions: bought recently, buy often, spend most
- * Loyal Customers: buy regularly, good spenders
- * At Risk: used to buy often but haven't recently
- * Lost: haven't bought in a long time
- * Can't Lose Them: big spenders who are becoming inactive
- * Recent Customers: bought recently but not often yet
- * Potential Loyalists: recent buyers with good frequency

WHY RFM? This is a real technique used by marketing teams to decide who to target with promotions and discounts.

VIEW 6: vw_payment_methods

- * Breakdown of revenue and transactions by payment method
- * Used for: understanding customer payment preferences

VIEW 7: vw_geo_revenue

- * Revenue and orders breakdown by city and state
- * Used for: identifying top markets geographically
- * Key insight: Delhi generates the highest revenue

8. DASHBOARD

We built TWO dashboards:

DASHBOARD 1 - STREAMLIT (Web-based, Python)

File: dashboard.py

Run with: `python -m streamlit run dashboard.py`

URL: `http://localhost:8501`

Features:

- * Dark theme modern design
- * 4 KPI cards at the top
- * Monthly revenue trend (line + bar combo)
- * Category revenue (donut chart)
- * Top products (horizontal bar)
- * Payment methods breakdown
- * RFM customer segments
- * Geographic revenue (city/state)

- * Cohort revenue analysis

DASHBOARD 2 - POWER BI (Interactive Business Dashboard)

File: dashboard.pbix

Features:

- * 4 KPI cards: Total Revenue, Total Orders, Total Customers, Avg Order Value
- * Monthly Revenue Trend (line chart)
- * Revenue by Category (donut chart)
- * Top Products by Revenue (bar chart)
- * Revenue by City (bar chart)
- * Payment Methods (column chart)
- * Customer Segments RFM (column chart)
- * Cross-filtering: click any chart to filter all others
- * Dark themed professional design

9. KEY BUSINESS INSIGHTS FROM THE DATA

REVENUE

- * Total Revenue: ~389 Million (3 years, 2022-2024)
- * Average Order Value: ~43,460
- * Revenue is consistent across all months (stable business)

PRODUCTS

- * Electronics dominates revenue (55.59% of total)
- * Smartphone X12 is the best selling single product
- * Books and Toys have lowest revenue contribution

CUSTOMERS

- * 2,000 total customers
- * Loyal Customers are the largest RFM segment
- * Delhi, Pune, and Chennai are top 3 cities by revenue

PAYMENTS

- * All payment methods (UPI, Credit Card, Debit Card, COD, Net Banking) are nearly equally popular
- * No single method dominates - healthy diversity

10. SKILLS DEMONSTRATED IN THIS PROJECT

DATA ENGINEERING

- * Designed normalized relational database schema
- * Built ETL pipeline from scratch using Python
- * Handled data cleaning, type conversion, deduplication

SQL / DATABASE

- * Created complex SQL views with JOINS and aggregations
- * Used window functions (NTILE) for RFM scoring
- * Used CTEs (WITH clauses) for cohort analysis
- * Designed foreign key relationships

PYTHON PROGRAMMING

- * File I/O (CSV reading and writing)

- * Database connectivity (mysql-connector-python)
- * Data generation with realistic randomization
- * Script automation (setup.py runs entire pipeline)

DATA VISUALIZATION

- * Built interactive dashboard with Streamlit + Plotly
- * Built business dashboard in Power BI
- * Connected Power BI to MySQL via ODBC
- * Created cross-filtering relationships in Power BI data model

BUSINESS ANALYTICS

- * RFM customer segmentation (real marketing technique)
- * Cohort analysis for customer retention
- * KPI tracking (Revenue, Orders, Customers, AOV)
- * Geographic and category performance analysis

11. HOW TO RUN THE PROJECT

REQUIREMENTS

- * Python 3.12+
- * MySQL Server 9.7+
- * Power BI Desktop

SETUP (first time only)

1. Open config.py and set your MySQL password
2. Run: `python setup.py`

This will create database, generate data, load data, and create all analytics views automatically.

RUN STREAMLIT DASHBOARD

- * Run: `python -m streamlit run dashboard.py`
- * Open browser: `http://localhost:8501`

OPEN POWER BI DASHBOARD

- * Open dashboard.pbix in Power BI Desktop
- * All charts will load automatically from MySQL

12. WHAT MAKES THIS PROJECT STRONG FOR A CV

1. END-TO-END: covers the full data analytics workflow (data ? database ? SQL ? visualization)
2. REAL TECHNIQUES: RFM segmentation and cohort analysis are used by actual companies (Amazon, Flipkart, etc.)
3. MULTIPLE TOOLS: Python + MySQL + Power BI shows versatility across the data analytics stack
4. BUSINESS CONTEXT: every chart answers a real business question - not just random charts
5. SCALE: 15,000 orders, 44,816 order items - realistic dataset size that demonstrates handling real data volumes

END OF PROJECT EXPLANATION

